BL-01 Decision Log - Requirement Conflicts and Attendance Rules

Backlog item: BL-01 (GitHub issue #1) | Sprint 1 | Decisions approved by the project team on 19 Oct 2026.

These decisions are recorded in the SRS (FR/NFR updates and rules AR-01 to AR-14), the Use Case Flow document, the Test Plan, and the Security Requirements Specification. If a document and this log ever disagree, raise it as a defect.

Decisions

ID	Topic	Decision	Reason	Recorded in
D-01	Student and faculty record permissions	Only the Admin adds, updates, deactivates, reactivates and permanently deletes student and faculty records. Faculty view students in their assigned classes only. Students view their own profile, read-only.	FR-03 allowed Faculty to add and delete students, but UC-02 named only the Admin. One accountable role protects student data.	FR-03, FR-04, AR-02, UC-02, UC-03, SEC-14
D-02	Account provisioning	Only the Admin creates accounts; no self-registration. Creating a student or faculty record creates the login with a one-time temporary password. Username = roll number (students) or faculty ID (faculty).	FR-01 said "registration" without defining it; SEC-06 already assumed Admin-created accounts.	FR-01, AR-01, UC-01, UC-02, UC-03
D-03	Deactivation and reactivation	Deactivation keeps all attendance and audit history, blocks login, ends sessions, removes the person from later attendance sheets, stops notifications, and labels past records "Inactive". The Admin can reactivate; a student reappears from the reactivation date and sessions held while deactivated are not counted. Faculty with active class assignments cannot be deactivated.	Keeps eligibility history while allowing a reversible removal.	AR-03, UC-02 5a-5b, UC-03 4a/4c, SEC-16
D-04	Permanent deletion	The Admin may permanently delete a student or faculty record after seeing what will be removed, confirming, and entering a reason. A student deletion removes the profile, login and that student's attendance. A faculty deletion keeps the attendance they recorded. Faculty with active assignments cannot be deleted. Audit log entries are never deleted.	The team needs a true delete (for example, records created in error) without losing accountability.	FR-03, FR-04, NFR-05, AR-04, UC-02 5c, UC-03 4b, SEC-16
D-05	Subject deletion	A subject with attendance records cannot be deleted.	Deleting it would erase many students' attendance. UC-04 4a said "warns or blocks".	FR-05, UC-04 4a
D-06	Attendance statuses	Present and Absent only. No late or partial credit; the faculty member decides for a late arrival.	FR-07 and UC-06 disagreed. Two statuses keep the calculation simple. A "Late" status can be proposed later as a new backlog item.	FR-07, AR-05, UC-06
D-07	Incomplete submissions	A submission with any unmarked student is rejected and nothing is saved. A "Mark remaining as Present" button fills only unmarked students, and only when clicked.	UC-06 4a allowed either rejecting or a silent default.	FR-07, AR-07, UC-06 4a
ID	**Topic**	**Decision**	**Reason**	**Recorded in**
D-08	Attendance session key	Section + subject + date + period number. One record per student per session.	No uniqueness rule existed; double periods (e.g. labs) must be possible.	FR-07, AR-06, UC-06 1b-1d

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		Duplicates are rejected. Same subject on one date is allowed in a different period. No future dates.		
D-09	Correction window and Admin override	Faculty may correct, or create a missed session, until 23:59:59 India Standard Time (IST, UTC+05:30) on the 2nd calendar day after the session date (example: session 6 Oct -> open until 8 Oct 23:59:59, blocked from 9 Oct 00:00). Default 2 days; the Admin can set 0-30. The Admin can correct any record, or record a missed session, at any time with a mandatory reason, logged as an Admin override.	SEC-23 only suggested 7 days; the team chose 2 days. Students who come after the deadline are handled by the Admin.	FR-08, FR-18, AR-08, UC-06 1c, UC-07 3a/3c, SEC-23
D-10	Percentage formulas and display	Subject % = attended / held x 100, shown as "10/12 (83.33%)". Overall % is pooled: total attended / total held (10/12 + 8/10 -> "18/22 (81.82%)"), not an average of subject percentages.	Showing counts lets students check the figure; pooling weights every class equally.	FR-09, FR-11, AR-09, AR-10, UC-08
D-11	Zero classes	Display "N/A - no classes yet"; never flagged or notified.	0 / 0 is undefined; 0% or 100% would mislead.	AR-11, AR-12, UC-08 4a, UC-10 2b
D-12	Precision and rounding	Store counts, not percentages. Display 2 decimals, rounding half up (2/3 -> 66.67%; 1/160 -> 0.63%).	Every screen and report shows the same figure.	AR-11
D-13	Threshold	Default 75%; the Admin can set a whole number 1-100. Flag when the exact value is below the threshold (attended x 100 < threshold x held). Equal is not flagged. Evaluated per subject and overall. No minimum number of sessions.	FR-13 said "below" without defining equality, rounding or scope.	FR-13, AR-12, UC-10
D-14	Student notifications	In-app notification and email, sent once each time a subject or overall percentage moves from not below to below the threshold. No repeats while below. The state resets on recovery (>= threshold). A threshold change re-evaluates everyone. If email fails, the in-app notification is still delivered and the failure is logged.	Prevents repeated alerts after every save; uses the in-app inbox and Email Provider in the component diagram.	FR-14, AR-13, UC-10, SEC-25
D-15	Faculty below-threshold list	Faculty and the Admin see an always-current in-app list (faculty: their assigned subjects only). Each faculty member receives a weekly email every Monday at 08:00, sent only when the list is not empty.	Faculty follow up with students without receiving every individual alert.	FR-14, AR-14, UC-10 step 5, SEC-25
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D-16	Quality targets	Performance: 95% of requests <= 2 s; submitting attendance for 60 students <= 3 s; one-semester section report <= 10 s; measured with 50 concurrent users. Capacity: >= 1,000 students, 60 faculty, 50 concurrent users, one academic year (about 750,000 records). Availability: 99% of 08:00-18:00, Mon-Sat, per month, excluding announced maintenance. Compatibility: latest two versions of Chrome, Edge, Firefox	NFR-01, 06, 08, 12 and 15 were not measurable. Targets sized for a single institution.	NFR-01, NFR-06, NFR-08, NFR-12, NFR-15

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		and Safari on desktop; usable at 360 px on Chrome (Android) and Safari (iOS). Accuracy: 100% of calculation tests pass.		
D-17	Security specification alignment	SEC-14 adds student records to Admin-only functions. SEC-16 also logs reactivation and permanent deletion with the reason. SEC-23 default changed from "suggested 7 days" to 2 calendar days, with Admin edits requiring a reason. SEC-25 sends faculty only the list of their own students. No other security requirement changed (15-minute idle timeout and 5-attempt lockout unchanged).	Avoids the security specification contradicting D-01, D-04, D-09 and D-15.	SEC-14, SEC-16, SEC-23, SEC-25 and their verification rows
D-18	Document format	SRS requirement tables use the columns Req ID, Type, Description, Priority, Acceptance Criteria, Rationale; priorities follow the project board (P0/P1/P2). UC-IDs added to the use case Goal rows. Use case diagram redrawn with associations matching the use case actors; editable source added (`use_flow_diagram.svg`). Component diagram labels updated: Notification Service includes the weekly faculty list, and the security note mentions the Admin override with reason.	Traceability between SRS, use cases, tests and backlog. The previous use case diagram linked Faculty to UC-05 instead of UC-04, linked Student to UC-09 and UC-11, omitted Student-UC-10 and Admin-UC-12, and contained unlabelled use-case-to-use-case lines.	SRS, Use*Case*Flow.docx, use*flow*diagram.pdf/.svg, Component_Diagram.drawio.png
D-19	Section enrolment	A student can be enrolled in only one section per subject at a time. Assigning a student to a second section for the same subject is rejected, and the message names the section the student is already in.	Otherwise the student appears on two attendance sheets for one subject and the percentage is wrong. Answers the open question in BL-10.	FR-06, UC-05 4a, ST*05*2

Decisions added later

ID	Topic	Decision	Reason	Recorded in
D-20	Two-factor login (BL-33, 26 Oct 2026)	After the password, every user - Admin, Faculty and Student - enters a 6-digit one-time code. Main method: an authenticator app (time-based codes, 30 seconds). Backup: a code sent by email, valid 10 minutes, single use, at most 3 requests per 15 minutes. No "remember this device".	Protects accounts if a password is guessed or stolen (TH-01, TH-02). Not in the original SRS; added by team decision.	SEC-29 to SEC-32, FR-21, UC-01, test plan UT*01*5-8, IT*01*4-5, ST*01*6-7
D-21	OTP setup and recovery (BL-33)	Users set up the authenticator app at first login (QR code plus one confirming code) and receive 10 single-use backup codes, shown once and stored only as hashes. An Admin can reset another user's OTP after re-entering their own password; the reset is logged and forces setup again.	Users who lose their phone can still log in without weakening security.	SEC-30, SEC-31, BL-05 permission matrix
D-22	OTP attempt limits (BL-33)	5 consecutive wrong codes lock the account for 15 minutes; every OTP setup, success, failure, lockout and reset is logged.	Same rule as passwords (SEC-03), so codes cannot be guessed.	SEC-32

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D-23	Forgotten passwords (6 Nov 2026)	No self-service reset links. **Forgot password?** on the login page sends a reset request: the other active Admins are emailed, the account shows "Reset requested", and the request is logged. The page gives the same reply for unknown usernames, and repeat requests within 15 minutes send no extra email. An Admin then resets the password; the new temporary password is emailed only to the account owner, who must change it at the next login and still completes two-factor login.	SEC-07 originally specified a single-use reset token expiring after 30 minutes. That conflicted with D-02 (Admin-controlled accounts) and was never built. This keeps account recovery under Admin control, and two-factor login (D-20) still protects the account if someone else reads the email. The temporary password has no time limit; it stays valid until used or replaced. Resolves VAL-01 in the Part 3 System Validation Report.	SEC-07 and its verification row, FR-01 criterion 5, UC-01 2a, test cases IT*01*6-7, UT*01*9-10, UT*01*13
D-24	Weekly timetable and class reminders (9 Nov 2026)	The Admin keeps a weekly timetable: each entry is one subject offering (subject in a section) on a weekday (Monday to Saturday) in a period (1 to 6). An entry is rejected if the section, or its teacher, already has a class at that time. Admin and Faculty dashboards list today's timetabled classes with their times, name the next class, and flag finished classes whose attendance is not marked; Mark attendance opens with the class, date and period filled in. Additions and removals are logged.	Faculty asked for a reminder of upcoming classes, and unmarked classes are the main cause of missing attendance. Not in the original SRS; added by team decision.	FR-22, UC-14, test cases UT*14*1-3, IT*14*1
D-25	Semester classes and attendance advice (9 Nov 2026)	The Admin may set how many classes each subject offering will hold this semester (1 to 500, optional; changes logged). Students see, per subject and overall, either how many of the remaining classes they must attend to finish at the threshold, or, when no total is set, how many classes in a row they must attend to get back to it (attended + x) / (held + x) >= threshold. If the threshold can no longer be reached, the best possible percentage is shown instead. The same advice appears in low-attendance notifications. Students can also open a subject to see every class marked Present or Absent.	Students asked how many more classes they need; a percentage alone does not answer that. Uses the existing formulas (D-10 to D-13); stored percentages are still never used.	FR-23, FR-09 criterion 4, FR-14 criterion 5, UC-08, UC-10, UC-14, test cases UT*14*4-7, IT*14*2-3
D-26	Send login details from the page (9 Nov 2026)	When a temporary password is shown (account created, password reset, or email failed), the Admin can enter or confirm the owner's address and send the details by email. Only the account's current temporary password can be sent (checked against the stored hash); a changed address is saved on the account and logged with a masked value; the send attempt and outcome are logged as in BL-34.	Accounts created without an email address could otherwise only receive their password by hand. Keeps D-23 (Admin-controlled recovery) and SEC-05 (password never stored readably).	FR-24, SEC-06, UC-01/UC-02/UC-03 addendum, test cases UT*01*15-17, IT*01*9
D-27	History lists records after filtering (9 Nov 2026)	Attendance history shows records once the filters are applied; applying with no filters lists every record in the user's scope. Record scope rules (SEC-13) are unchanged.	Requested during review so that the page opens on the filters rather than a long unfiltered list.	FR-10 criterion 4, UC-11, test case IT*11*4

ID	Topic	Decision	Reason	Recorded in
D-28	Dashboard counts and interface redesign (9 Nov 2026)	Shortage counts on dashboards exclude deactivated students, matching the below-threshold list and alerts (D-03, D-15), and the summary appears only when someone is below the threshold. The interface gains a logo, a role-based sidebar, a split sign-in page and buttons in place of text links; each page keeps one Back to dashboard button. No permission changes: the sidebar shows only pages the role may open, and the server still checks every request (SEC-10).	A dashboard said one student was below the threshold while the list it opened showed none. The redesign came from usability review (NFR-07).	FR-17 criteria 4-5, NFR-07, UC-12, test cases IT*12*5-6

Confirmations (19 Oct 2026)

- Threshold: 75% confirmed as the default (D-13); the Admin can change it within 1-100 as required by SEC-14, UC-10 and BL-22.
- Time zone for the correction window: IST (D-09).
- Backlog items BL-12 (#12), BL-17 (#17) and BL-24 (#24) updated on GitHub to match D-08, D-09 and D-15.

Open items

None.